Auditing the Costs of Multi-Group Fairness Repair

XIN ZHENG and SHIHAN FENG

Equal opportunity (EO) requires the same true-positive rate (TPR) across demographic subgroups. Enforcing EO for each sensitive attribute separately does not ensure EO across their intersections. We audit three existing methods for intersectional EO on Credit Default and COMPAS, using logistic regression and histogram gradient boosting. We examine accuracy loss and withdrawals of previously favorable predictions, both overall and within subgroups, and compare repair-set and test-set outcomes. All three methods achieve near-zero intersectional TPR gaps on the repair set, but substantial test gaps remain on COMPAS, where accuracy losses and withdrawals are also large. We then introduce Retain-EG, which adjusts the selection probabilities of classifiers learned by exponentiated gradient (EG) to minimize withdrawals under explicit TPR-gap and accuracy constraints. Retain-EG reduces test withdrawals across all evaluated models and splits, with nearly unchanged average accuracy. However, it does not consistently improve TPR gaps, accuracy, and withdrawals together, and some subgroups still lose accuracy or experience more withdrawals. These results show why the costs of fairness repair and its performance on new data must be evaluated together.

Additional Key Words and Phrases: equal opportunity, intersectional fairness, postprocessing, decision withdrawals

1 Introduction

Equal opportunity (EO) requires the same true-positive rate (TPR) across demographic subgroups: samples with a favorable true label should have the same probability of a favorable prediction regardless of subgroup membership. Here, a subgroup can be defined by one sensitive attribute, such as sex, or by a combination of attributes, such as race and sex. We call comparisons based on one attribute *marginal* and comparisons based on a combination of attributes *intersectional*. Enforcing marginal EO does not necessarily ensure intersectional EO.

For a trained classifier, EO can be pursued through postprocessing, which changes decision thresholds or the probabilities of favorable predictions. These changes can affect accuracy and reverse previously favorable predictions. A withdrawal changes an originally favorable prediction to an unfavorable one. In applications such as lending, this can remove an opportunity that the original classifier would have granted. Measuring withdrawals records how often such opportunities are removed.

Consider eight applicants who all have favorable true labels, divided into two subgroups of four, as illustrated in Figure 1. The original classifier approves three applicants in one subgroup and one in the other. Two repairs each approve two applicants per subgroup, achieving EO with identical TPRs and accuracy. The first withdraws one original approval and approves one previously rejected applicant. The second withdraws two original approvals and approves two previously rejected applicants. Their withdrawal rates are therefore 25% and 50%. Although both repairs meet the EO requirement, the second denies an additional originally approved applicant without improving either TPRs or accuracy.

We audit Hardt–Price–Srebro postprocessing (Hardt) [3], Morina et al.'s Randomized Thresholding Derived Predictor (RTDP) [5], and Agarwal et al.'s Exponentiated Gradient (EG) reduction [1]. On Credit Default and COMPAS, the methods use the same scores from logistic regression (LR) and histogram gradient boosting (HGB). We compare marginal and intersectional TPR gaps, accuracy loss, and withdrawals on repair and test data. Accuracy and withdrawals are evaluated both overall and within subgroups. The audit examines which costs accompany fitted EO, how they are distributed, and whether the repair generalizes.

Building on the audit, we propose Retain-EG, which adjusts the selection probabilities of EG's existing classifiers to minimize withdrawals while constraining TPR gaps and accuracy on repair and validation data. It reduces test

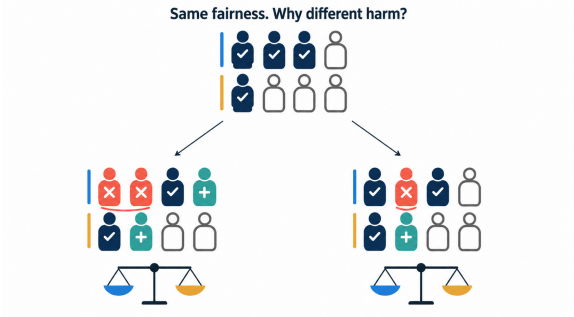


Fig. 1. Illustrative Example

withdrawals across both models and all five splits of each dataset, with nearly unchanged average accuracy. Its test TPR gaps do not improve consistently, and some subgroups still lose accuracy or experience more withdrawals. Code is available at https://github.com/Xin-Lyra-Zheng/DS-709-Final-Project.

2 Related Work

Fairness across individual sensitive attributes does not necessarily protect their intersections. Kearns et al. [4] demonstrate this limitation and develop methods for protecting richer sets of subgroups. We focus on the intersections defined by the sensitive attributes in each dataset.

Hardt et al. [3] introduce postprocessing for EO and equalized odds. Morina et al. [5] address intersectional fairness through subgroup-specific thresholds and randomized predictions. Agarwal et al. [1] reduce fairness-constrained classification to cost-sensitive learning and produce a distribution over classifiers. These methods provide our baselines. Cotter et al. [2] study the generalization of data-dependent constraints using separate samples for model and constraint learning, motivating our use of validation data when modifying a fitted repair.

Field Landscape. Within AI ethics, algorithmic fairness is part of a broader concern with how models are evaluated and deployed. EO specifies a comparison between subgroups, while evaluating a system's effects also requires asking who benefits and who bears its costs. Our audit connects these questions by examining accuracy and changes to original approvals across subgroups. It also relates to evaluating models after deployment, because fairness measured during development may not persist on new data.

Gap Analysis. Accuracy alone does not identify withdrawals: repairs with the same TPRs and accuracy can reverse different samples' predictions. We therefore audit accuracy loss and withdrawals as distinct costs under an EO requirement, across subgroups and on both repair and test data. Retain-EG then tests whether withdrawals can be reduced under explicit TPR-gap and accuracy constraints.

3 Methods

3.1 Baseline methods

Appendix A summarizes the notation used throughout the paper. We evaluate each combination of dataset, prediction model, and random data split separately, and call one such combination a fit. Let R, V, and T be the repair, validation, and test sets. For sample i, let x_i be its features, $y_i \in \{0, 1\}$ its label, and g_i its intersectional subgroup. The favorable

label is $y_i = 1$. Let $s(x)$ be the fixed prediction score. We use q consistently for favorable-prediction probabilities: $q^{\text{orig}}(x, g) \in \{0, 1\}$ denotes the original classifier's deterministic prediction, and $q(x, g) \in [0, 1]$ denotes a repair's probability of a favorable prediction. Their values for sample i are $q_i^{\text{orig}} = q^{\text{orig}}(x_i, g_i)$ and $q_i = q(x_i, g_i)$. A threshold rule gives a favorable prediction when the score is at or above its threshold.

For any of these sets D and any evaluated subgroup g, let D_g contain the samples belonging to g, and let $D_g^+ = \{i \in D_g : y_i = 1\}$. Subgroups can be marginal or intersectional, as defined in the Introduction. Write $|S|$ for the number of samples in a set S. The TPR and accuracy are

$$\widehat{\text{TPR}}_{g,D}(q) = \frac{1}{|D_g^+|} \sum_{i \in D_g^+} q_i, \tag{1}$$

$$\widehat{\text{Acc}}_{g,D}(q) = \frac{1}{|D_g|} \sum_{i \in D_g} \left[y_i q_i + (1 - y_i)(1 - q_i) \right]. \tag{2}$$

Both quantities average over prediction randomization.[1] Overall accuracy, $\widehat{\text{Acc}}_D(q)$, uses D in place of D_g. TPR is defined when $|D_g^+| > 0$.

Let $\mathcal{G}_{\text{fit}}$ be the intersectional subgroups included in a baseline's EO constraints, each containing positive-label repair samples. We distinguish this fitting set from the sets used in the reported comparisons below. Using g and g' for two subgroups, EO on the repair set requires

$$\widehat{\text{TPR}}_{g,R}(q) = \widehat{\text{TPR}}_{g',R}(q) \qquad \forall g, g' \in \mathcal{G}_{\text{fit}}. \tag{3}$$

Hardt–Price–Srebro postprocessing. Hardt et al. [3] choose threshold rules separately for each subgroup and randomize between them to maximize accuracy subject to EO. For example, choosing with equal probability between rules with TPRs of 40% and 80% gives an expected TPR of 60%. The method chooses the rules and their selection probabilities so that the subgroups have equal repair-set TPRs.

Randomized Thresholding Derived Predictor (RTDP).. Morina et al. [5] set one threshold for each subgroup and assign two probabilities of a favorable prediction: one for samples whose scores are at or above the threshold, and another for samples whose scores are below it. We use their sequential procedure: first choose each threshold to maximize its subgroup's repair-set accuracy, then hold the thresholds fixed and optimize the two probabilities per subgroup to maximize overall accuracy subject to EO. Samples in the same subgroup and on the same side of its threshold receive the same probability. Fixing thresholds before enforcing EO need not yield the best joint choice of thresholds and probabilities.

Exponentiated Gradient (EG).. Agarwal et al. [1] maximize accuracy subject to fairness constraints by alternating between classifier search and constraint updates. Let K be the number of constraints, $\widehat{\gamma}_k(q)$ the TPR difference in constraint k, and d_k its permitted upper bound. Let $\lambda = (\lambda_1, \ldots, \lambda_K)$ be the vector of nonnegative multipliers, with λ_k controlling the influence of constraint k:

$$L(q, \lambda) = 1 - \widehat{\text{Acc}}_R(q) + \sum_{k=1}^{K} \lambda_k \big(\widehat{\gamma}_k(q) - d_k \big), \tag{4}$$

[1] A positive-label sample is predicted correctly with probability q_i, and a negative-label sample with probability $1 - q_i$. Accuracy averages these probabilities rather than the results of one random draw of binary predictions.

This objective, called the Lagrangian, combines classification error with fairness constraints: exceeding a TPR-difference bound increases the objective, with a larger multiplier making that violation more costly.

For fixed λ, the objective assigns costs to predicting 0 or 1 for each repair sample. Finding a classifier with the lowest total cost is called cost-sensitive classification. The procedure performing this search is called an oracle. Our oracle searches subgroup-specific threshold rules over the fixed prediction scores. EG then updates the multipliers using exponentiated gradient, a multiplicative update that gives relatively more emphasis to more strongly violated constraints, and repeats the search.

Let $h_1, \ldots, h_M$ be the M returned binary classifiers, with $h_t(x, g) \in \{0, 1\}$ denoting classifier t's prediction. EG combines them because randomizing between classifiers can balance their different subgroup TPRs. Let $w = (w_1, \ldots, w_M)$ be their selection probabilities. The probability simplex $\Delta_M = \{w \in \mathbb{R}^M : w_t \geq 0 \; \forall t = 1, \ldots, M, \; \sum_{t=1}^{M} w_t = 1\}$ contains all permitted probability vectors. The probability of a favorable prediction is

$$q_w(x, g) = \sum_{t=1}^{M} w_t h_t(x, g). \tag{5}$$

We save the classifiers and EG's final selection probabilities w^{EG} for Retain-EG, which changes these probabilities while keeping the classifiers fixed.

3.2 Evaluation metrics

We audit TPR gaps, accuracy loss, and withdrawals on R and T, using the same original predictions q^{orig} as the reference for every repair. Equations (1) and (2) apply to both the original predictions and the repaired probabilities.

TPR gap. Let $\mathcal{G}_j$ be the subgroup set for comparison $j = 1, \ldots, J$. Here $J = 3$: $\mathcal{G}_1$ contains sex subgroups. $\mathcal{G}_2$ contains age subgroups for Credit and race subgroups for COMPAS. and $\mathcal{G}_3$ contains the reported intersectional subgroups selected using the cutoff in Section 4.1. Thus, $j = 1, 2$ are marginal comparisons and $j = J = 3$ is the intersectional comparison. The TPR gap for comparison j on set D is

$$\widehat{\mathrm{Gap}}_{j,D}(q) = \max_{g \in \mathcal{G}_j} \widehat{\mathrm{TPR}}_{g,D}(q) - \min_{g \in \mathcal{G}_j} \widehat{\mathrm{TPR}}_{g,D}(q). \tag{6}$$

We report the comparisons separately to assess EO for each sensitive attribute and their intersections, and its generalization from repair to test data.

Accuracy loss. Accuracy loss compares a repair with the original predictions on the same samples:

$$\Delta \mathrm{Acc}_{g,D}(q) = \widehat{\mathrm{Acc}}_{g,D}(q^{\mathrm{orig}}) - \widehat{\mathrm{Acc}}_{g,D}(q). \tag{7}$$

The overall loss $\Delta \mathrm{Acc}_D(q)$ uses D in place of D_g. A negative loss means improved accuracy. Worst-group accuracy is $\min_{g \in \mathcal{G}_J} \widehat{\mathrm{Acc}}_{g,D}(q)$. It measures the lowest final accuracy, rather than the largest loss from a subgroup's original accuracy.

Withdrawals. The withdrawal rate is the fraction of originally favorable predictions reversed by a repair:

$$\widehat{W}_{g,D}(q) = \frac{\sum_{i \in D_g} q_i^{\mathrm{orig}} (1 - q_i)}{\sum_{i \in D_g} q_i^{\mathrm{orig}}}. \tag{8}$$

This rate is defined when the denominator is positive. The overall rate $\widehat{W}_D(q)$ uses all samples in D, and the maximum-group rate is $\max_{g \in \mathcal{G}_J} \widehat{W}_{g,D}(q)$. Withdrawals condition on originally favorable predictions, including incorrect ones, whereas the false-negative rate conditions on positive labels. Withdrawing a correct favorable prediction lowers accuracy while withdrawing an incorrect one improves it.

All metrics average over prediction randomization. Repair-to-test differences therefore reflect differences between samples and the fitted repair's performance on new data, rather than additional variation from drawing binary predictions. Worst-group accuracy and maximum-group withdrawal are calculated within each fit before averaging.

3.3 Retain-EG

Different selection probabilities can produce similar TPR gaps and accuracy while withdrawing different originally favorable predictions. Retain-EG keeps EG's classifiers fixed and changes only their selection probabilities, subject to TPR-gap and accuracy constraints on R and V.

Let ϵ be the TPR-gap tolerance, δ the maximum accuracy decline relative to EG, and ρ the maximum total change in selection probabilities. We use $\epsilon = 0.02$, $\delta = 0.005$, and $\rho = 0.25$. For each comparison $j = 1, \ldots, J$ and set $D \in \{R, V\}$, the permitted gap is

$$\kappa_{j,D} = \max\{\epsilon, \widehat{\mathrm{Gap}}_{j,D}(q_{w^{\mathrm{EG}}})\}. \tag{9}$$

Thus, an EG gap below two percentage points may increase to two points, while a larger EG gap cannot increase. Including EG's existing gap in the bound ensures that its saved probabilities w^{EG} satisfy the constraint.

We measure the total change in selection probabilities by $\|w - w^{\mathrm{EG}}\|_1 = \sum_{t=1}^{M} |w_t - w_t^{\mathrm{EG}}|$ and solve

$$\begin{aligned} \min_{w \in \Delta_M} \quad & \widehat{W}_R(q_w) \\ \text{subject to} \quad & \widehat{\mathrm{Gap}}_{j,D}(q_w) \le \kappa_{j,D} \quad \forall j = 1, \ldots, J,\ D \in \{R, V\}, \\ & \widehat{\mathrm{Acc}}_D(q_w) \ge \widehat{\mathrm{Acc}}_D(q_{w^{\mathrm{EG}}}) - \delta \quad \forall D \in \{R, V\}, \\ & \|w - w^{\mathrm{EG}}\|_1 \le \rho. \end{aligned} \tag{10}$$

The objective minimizes overall repair-set withdrawals. The constraints bound TPR gaps, allow at most a half-percentage-point accuracy decline, and limit changes to EG's classifier selection probabilities.

Accuracy and withdrawals are linear in w. A gap bound is equivalent to bounding every pairwise subgroup TPR difference, which also gives linear constraints. The problem is therefore a linear program. Since w^{EG} satisfies all constraints, optimized repair-set withdrawals cannot increase except for numerical tolerance. This does not guarantee test-set improvement.

4 Experiments

4.1 Data and experimental design

We use UCI Credit Default [7], abbreviated as Credit, and COMPAS [6]. Credit contains 30,000 samples. The favorable label of the target is no default in the following month. Its sensitive attributes are sex and age below 40 versus at least 40. COMPAS contains 6,172 samples after filtering. The favorable label of the target is no recorded recidivism within two years. Its sensitive attributes are race and sex.

For each dataset, five random seeds create disjoint training, calibration, repair, validation, and test sets in proportions 50/10/15/10/15. We train logistic regression (LR) and histogram gradient boosting (HGB), use the calibration set to map

their scores through fitted sigmoid functions, and then fix the calibrated scores. For each model, the original classifier's threshold maximizes repair-set accuracy. The baselines use the same scores, original predictions, and data splits. We fit the baselines on R, use both R and V to constrain Retain-EG, and evaluate final outcomes on T.

Our reported intersectional comparisons include only subgroups with at least twenty positive-label repair samples. TPR is estimated from these samples, so this cutoff avoids very small denominators. It retains four Credit subgroups and six COMPAS subgroups in every fit. Appendix **??** provides partition sizes, subgroup definitions, filtering, and baseline settings.

4.2 Results

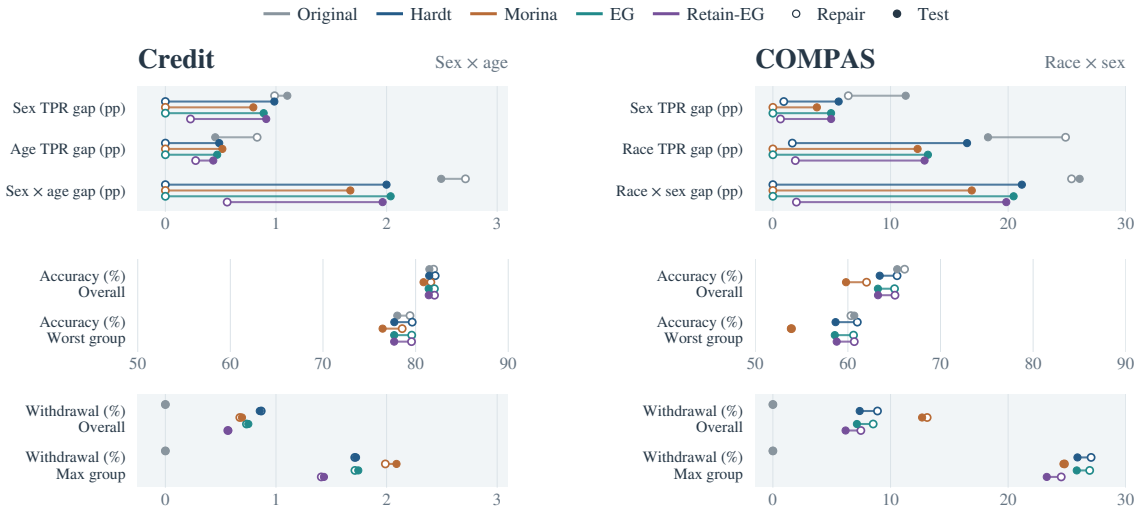


Fig. 2. TPR gaps, accuracy, and withdrawals on Credit and COMPAS. Open and filled points show repair-set and test-set means over two models and five splits. Lines connect results for the two sets. Accuracy is reported directly and withdrawal rates are percentages of original approvals. Worst-group accuracy and maximum-group withdrawal are calculated within each fit before averaging. Each metric has its own axis.

4.2.1 Baseline audit. Figure **??** compares TPR gaps, accuracy, and withdrawals on repair and test data. All three baselines achieve approximately zero intersectional repair-set TPR gaps, but their test performance differs across datasets. Interestingly, on Credit, test gaps remain small for individual sensitive attributes but are larger for their intersections. Accuracy changes and withdrawals are also small. In contrast, on COMPAS, substantial test gaps remain for both individual attributes and their intersections, accompanied by larger accuracy losses and more withdrawals. For these three repairing methdos, reporting only repair-set TPR gaps would conceal both remaining test disparities and the costs of achieving fitted EO.

Looking into the details on COMPAS, RTDP achieves the smallest mean test intersectional TPR gap but incurs the largest overall accuracy loss and withdrawal rate. Hardt and EG retain more accuracy and original approvals while leaving larger test gaps. The method with the highest overall withdrawal rate does not have the highest maximum-group rate. Therefore, all three baselines have their own deficits. Furthermore, for all three methods, worst-group accuracy

declines and maximum-group withdrawal rates substantially exceed overall rates. Overall averages therefore give an incomplete picture of how the costs are distributed and hides the real unfairness across subgroups.

4.2.2 Retain-EG. Relative to EG, Retain-EG reduces test withdrawals in all twenty fits, with nearly unchanged average accuracy and smaller mean intersectional TPR gaps (Table 1). However, TPR gaps, accuracy, and withdrawals all improve or remain unchanged in only half the fits on each dataset. The mean improvements therefore do not describe every fit.

Table 1. Test outcomes of Retain-EG relative to EG, averaged over two models and five splits per dataset. Withdrawal rates are percentages of original approvals. Changes subtract EG's result from Retain-EG's result: a negative TPR-gap change or positive accuracy change indicates improvement. Both changes are in percentage points (pp). The final column counts fits with no larger intersectional TPR gap, no lower accuracy, and no greater withdrawal rate.

Dataset	EG withdrawal	Retain-EG withdrawal	TPR-gap change (pp)	Accuracy change (pp)	All three improve or tie
Credit	0.750%	0.569%	−0.07	≈ +0.01	5/10
COMPAS	7.144%	6.181%	−0.62	≈ +0.01	5/10

In Figure 2, subgroup outcomes also vary. Some subgroups lose accuracy relative to EG, and one Credit subgroup experiences more withdrawals in one fit despite the overall reduction. Retaining more original approvals preserves both correct and incorrect predictions, so fewer withdrawals need not improve accuracy.

5 Discussion

Achieving EO on the repair set does not tell us how a method will perform on the test set or what accuracy and withdrawal costs it will incur. A smaller TPR gap may accompany larger accuracy losses or more withdrawals. Accuracy alone also cannot identify withdrawals, since repairs with the same accuracy can reverse different samples' predictions. Evaluating all three outcomes is therefore necessary to understand what a fairness repair achieves and what it costs.

Retain-EG shows that changing the selection probabilities of EG's classifiers can reduce withdrawals while keeping average accuracy nearly unchanged. However, its constraints permit some increase in repair-set and validation-set TPR gaps. The withdrawal reduction does not establish improvement at the same TPR gap. Nor do the overall improvements apply to every subgroup: minimizing overall withdrawals and constraining overall accuracy do not protect each subgroup separately.

The experiments have limitations. TPR estimates depend on the number of positive-label samples, and the reported intersectional comparisons exclude subgroups with fewer than twenty such repair samples. The same samples recur across splits, so the twenty fits are not independent tests. Test results were also inspected during development, making evaluation on new data necessary to confirm the findings. Finally, the experiments measure hypothetical prediction changes, rather than their effects on people in an actual lending or criminal justice system.

6 Conclusion

Near-zero repair-set TPR gaps can coexist with substantial test disparities, accuracy losses, and withdrawals that fall unevenly across subgroups. Retain-EG reduces test withdrawals with nearly unchanged average accuracy, but does not consistently improve all three outcomes or protect every subgroup. Intersectional EO should therefore be evaluated alongside accuracy and withdrawals, both overall and within subgroups, on repair and test data.

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A Notation Summary

The same notation is used in the main text and appendices. Hats indicate empirical quantities calculated on a specified data set. All favorable-prediction probabilities use q; superscripts and subscripts identify the original classifier, a sample, or a classifier distribution. The fitting subgroups $\mathcal{G}_{\text{fit}}$ and the reported subgroups $\mathcal{G}_j$ are kept distinct because their inclusion criteria can differ.

Table 2. Notation used throughout the paper.

Symbol	Meaning
$R,\ V,\ T$	Repair, validation, and test sets.
D	A generic data set, chosen from R, V, T; Retain-EG constraints use $D \in \{R, V\}$.
$i,\ x_i,\ y_i$	Sample index, features, and binary label. The favorable label is $y_i = 1$.
g_i	Sample i's intersectional subgroup, defined by its sensitive attributes.
$g,\ g'$	Subgroups in a marginal or intersectional comparison. The two indices denote any pair of subgroups.
$D_g,\ D_g^+$	Samples in D belonging to subgroup g, and the subset with $y_i = 1$, respectively.
$\lvert S\rvert$	Number of samples in a set S.
$\mathcal{G}_{\text{fit}}$	Intersectional subgroups included in a baseline's fitting constraints. Inclusion can depend on the method's label requirements.
$j,\ J$	Index and number of reported subgroup comparisons. Here $J = 3$: two marginal comparisons and one intersectional comparison.
$\mathcal{G}_j$	Subgroup set for comparison j; $\mathcal{G}_J$ is the reported intersectional set.
$s(x)$	Fixed prediction score for features x, supplied by LR or HGB.
$q^{\text{orig}}(x, g)$	Original classifier's deterministic prediction, viewed as a favorable-prediction probability in $\{0, 1\}$.
q_i^{orig}	Original prediction for sample i: $q^{\text{orig}}(x_i, g_i)$.
$q(x, g),\ q_i$	A repair's favorable-prediction probability in $[0, 1]$, and its value $q(x_i, g_i)$ for sample i.
$t,\ M$	Classifier index and number of saved EG classifiers.
$h_t(x, g)$	Saved classifier t's binary prediction, taking values in $\{0, 1\}$.
$w,\ w_t$	Vector of classifier selection probabilities and the probability of selecting classifier t.
$w^{\text{EG}},\ w_t^{\text{EG}}$	EG's saved selection-probability vector and its component for classifier t; Retain-EG uses this vector as its reference.
Δ_M	Probability simplex: nonnegative vectors of M selection probabilities summing to one.
$q_w(x, g)$	Favorable-prediction probability from the saved classifiers with probabilities w: $\sum_{t=1}^{M} w_t h_t(x, g)$.
$q_{w^{\text{EG}}}$	EG's fitted prediction probability; distinct from the original classifier q^{orig}.
$k,\ K$	Index and number of constraints in the EG reduction.
$\widehat{\gamma}_k(q),\ d_k$	TPR difference in EG constraint k and its permitted upper bound.

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Table 2 continued

Symbol	Meaning
$\lambda,\ \lambda_k$	Vector of nonnegative EG constraint multipliers and the multiplier for constraint k.
$L(q, \lambda)$	Lagrangian: classification error plus the weighted EG constraint terms.
$\widehat{\mathrm{TPR}}_{g,D}(q)$	Subgroup g's mean favorable-prediction probability among positive-label samples in D.
$\widehat{\mathrm{Acc}}_{g,D}(q),\ \widehat{\mathrm{Acc}}_D(q)$	Accuracy within subgroup g and overall, averaged over prediction randomization.
$\widehat{\mathrm{Gap}}_{j,D}(q)$	Largest minus smallest subgroup TPR over $\mathcal{G}_j$ on D.
$\Delta\,\mathrm{Acc}_{g,D}(q),\ \Delta\,\mathrm{Acc}_D(q)$	Accuracy loss relative to q^{orig}, within subgroup g and overall. Positive values indicate a loss.
$\widehat{W}_{g,D}(q),\ \widehat{W}_D(q)$	Withdrawal rate among originally favorable predictions, within subgroup g and overall.
$\kappa_{j,D}$	Retain-EG's permitted TPR gap for comparison j on D: $\max\{\epsilon, \widehat{\mathrm{Gap}}_{j,D}(q_{w^{\mathrm{EG}}})\}$.
ϵ	Retain-EG's TPR-gap tolerance, set to 0.02.
δ	Permitted overall accuracy decline relative to EG on each constrained set, set to 0.005.
ρ	Permitted total absolute change in classifier selection probabilities, set to 0.25.
$\lVert w - w^{\mathrm{EG}} \rVert_1$	Total absolute change in selection probabilities: $\sum_{t=1}^{M} \lvert w_t - w_t^{\mathrm{EG}} \rvert$.